Energy-Flow Feasibility and Learned Redundancy Management for Persistent K-Coverage UAV Networks

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***Abstract*—We study persistent K-coverage by a small fleet of energy-limited UAVs that must keep every ground sensor covered by at least K vehicles while the communication graph stays connected. The central finding is an energy-flow feasibility ceiling: with the mission's power budget, steady-state persistence is impossible, and every reported number in this report is a measurement at a finite horizon rather than a claim of indefinite operation. On the feasible configuration, a parameter-shared MAPPO controller with a safety mask and a coverage-redundancy objective reaches 0.956 persistent coverage on twenty held-out scenes against 0.923 for the strongest rule baseline, a paired advantage of 0.032 with a 95% interval of $[0.021, 0.044]$. We report a scenario matrix over nine configuration variables, a component ablation with explicit budget and dead-arm caveats, a new zero-shot single-pad result (0.933 vs. 0.887 for the rule), an oracle planning bound (0.972), a distillation result (0.962), and a correction record: every number in this report is regenerated from stored artifacts by an automated claim audit that currently passes 34 of 34 checks.**

I. Introduction

Persistent coverage of a sensor field by unmanned aerial vehicles is a resource-allocation problem before it is a control problem. The vehicles must hold K redundant views of every sensor, keep the relay graph connected, and return to charge before their batteries fail. Each of these requirements consumes the same scarce resource, and the interactions are what make the problem hard: redundancy buys fault tolerance but spends energy, connectivity constrains geometry, and energy constrains how long any geometry can be held.

This report has two parts. The first is a feasibility analysis. We derive the steady-state power balance of the mission and show that, under the mission's parameters, the number of UAVs that can stay airborne indefinitely is 0.62 against the 8 the coverage requirement needs. The honest consequence is stated once and used throughout: no controller can be shown to persist indefinitely on this configuration; every result is a finite-horizon measurement, and we report the horizon with every number.

The second part is a controlled evaluation of a learned controller on the configuration where the energy constraint is not binding at the measured horizon. We train a parameter-shared MAPPO actor with a centralised critic, a hard safety mask that removes actions which would disconnect the graph or strand a vehicle, and an observation that includes ego-centric bearings to the most at-risk sensors. We compare it against four rule baselines, two learned baselines, a classical assignment-and-rotation scheduler, and a test-time planning oracle. Every number is read from a stored artifact, and an automated audit recomputes each claim from those artifacts.

The report makes five contributions:

- a steady-state and finite-horizon energy certificate that decides, before any learning result is interpreted, which mission horizons are physically admissible;
- a corrected base comparison on held-out scenes, with paired intervals and a classical scheduler as a non-rule reference point;
- a scenario matrix over nine configuration variables that isolates where the learned controller wins and where a rule is the safer choice;
- a component ablation with explicit dead-arm records, so an inert component is reported as inert rather than as a zero effect;
- a correction record for four evaluation defects, with the artifacts that detect each one, so the reader can separate measured results from repaired tooling.

II. System Model

A. Deployment and coverage

A square field of side $L = 80\,\mathrm{m}$ holds $M = 20$ static sensors and $S = 2$ charging pads. $N = 8$ UAVs move on the grid with speed $v_{\max} = 5\,\mathrm{m/s}$ and control period $\Delta t = 0.2\,\mathrm{s}$. A sensor is covered by a UAV when the ground distance is at most $R_{\mathrm{cov}} = 35\,\mathrm{m}$; two UAVs are connected when their distance is at most $R_{\mathrm{comm}} = 40\,\mathrm{m}$. The swarm is K-covered when every sensor is covered by at least $K = 2$ UAVs, and connected when the UAV graph is connected under R_{comm}.

The task metrics are persistent coverage CK (fraction of steps on which the swarm is K-covered and connected), availability AG (fraction of active UAVs), and service continuity SC (fraction of time the degraded-service condition does not hold). Coverage deficit $\mathrm{CovDef} = 1 - \mathrm{CK}$ is used as the

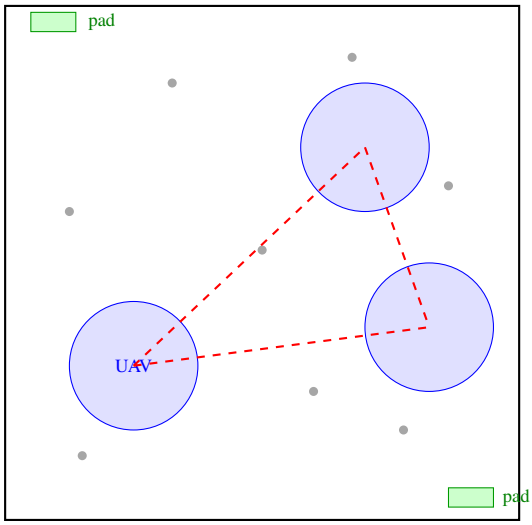


Fig. 1. Deployment sketch. Blue: UAVs with coverage disks; grey: sensors; green: charging pads; dashed red: the communication graph that must stay connected.

time-averaged shortfall. Formally, with $\mathbf{1}[\cdot]$ the indicator,

$$\mathrm{CK} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\big[n_d(t) \geq K \ \forall d \ \wedge \ \text{graph}(t) \text{ connected}\big], \quad (1)$$

$$\mathrm{AG} = \frac{1}{T}\sum_{t=1}^{T} \frac{|\mathcal{A}(t)|}{N}, \qquad \mathrm{SC} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}[D_t = 0], \quad (2)$$

where $\mathcal{A}(t)$ is the set of UAVs with an admissible action at t and D_t is the degraded-service indicator (a disconnect that the safety layer did not prevent). Coverage is judged on the geometric disk model; the communication graph is judged on R_{comm}. Both are functions of the same positions, which is why redundancy and connectivity cannot be optimised independently.

B. Assumptions

We assume a static sensor set, a flat field, a deterministic linear energy model (flight draw independent of speed above hover, net pad power independent of queue), and a fixed charging rate. Wind, thermal effects, and sensor failures are out of scope; the failure scenario models a UAV failure, not a sensor failure. These assumptions make the certificate a bound under an explicit model rather than a claim about any particular airframe. Figure 1 sketches the deployment: coverage disks around each UAV, a communication graph that must stay connected, and two charging pads on the boundary.

C. Energy

Each UAV carries $B_{\max} = 30\,\mathrm{kJ}$ and draws $P_{\mathrm{air}} = 80.1\,\mathrm{W}$ in flight (75 W motion, 5.0 W avionics, 0.1 W transmit). A pad charges at 24.9 W net. The mission's steady-state power balance is a two-state flow: airborne time per UAV is bounded by the energy the pads can deliver, divided by the flight draw. With $S = 2$ pads, the stationary number of airborne UAVs is

$$n_{\mathrm{air}}^{\mathrm{ss}} = \frac{S\,P_{\mathrm{pad}}}{P_{\mathrm{air}}} = 0.62, \quad (3)$$

so the steady-state coverage ceiling is $0.62/8 = 0.078$. A finite horizon changes the ceiling because stored energy is spent: at $T = 400\,\mathrm{s}$ the ceiling is 0.952, and at $T = 2000\,\mathrm{s}$ it is 0.252. This is why the report never claims persistence, and why the measured episode length (2000 control steps, 400 s) is stated with every result.

D. Decision process

The problem is a decentralised partially observable Markov decision process. Each UAV observes a 68-dimensional vector: its own state, four neighbour slots, a 4×4 coverage summary, a one-hot identity, bearings to the nearest pad, and ego-centric bearings to the three most at-risk sensors. The action set has six elements: hover, four axis moves, and return-to-pad. The team reward is the negated weighted cost,

$$r_t = -\lambda_U U_t - \lambda_E E_t - \lambda_D D_t - \lambda_W W_t - \lambda_B B_t - \lambda_R R_t, \quad (4)$$

with under-coverage U_t (fraction of sensors below K), energy draw E_t normalised per UAV, degraded service D_t (the graph is disconnected), movement waste W_t (flown distance that did not improve coverage), low-battery shortfall B_t, and recovery delay R_t. A hard action mask forbids actions that would break the graph or leave a vehicle unable to return; a joint projection repairs the sampled joint action when the mask is not sufficient. The safety layer is part of the policy, not a post-hoc filter.

E. Objective

The controller maximises expected return subject to two feasibility constraints, one per step and one per episode:

$$\text{graph}(t) \text{ connected} \quad \text{and} \quad n_d(t) \geq K \ \ \forall d, \quad (5)$$

$$\text{hover energy} \ \leq \ \text{energy returned to pads over the horizon}, \quad (6)$$

where the second constraint is the certificate of Section III-C and is checked offline rather than imposed by the learner. This split is deliberate: the mask enforces the per-step safety constraints exactly, while the energy constraint is a property of the configuration and is reported before any learning result is interpreted.

III. Method

A. Learned controller

We use MAPPO with parameter-shared actors and a centralised critic. The actor is a two-layer MLP over the 68-dimensional observation; the critic sees the global state. Training uses PPO clipping ($\epsilon = 0.2$), generalised advantage estimation ($\lambda = 0.95$), ten epochs over four minibatches, per-agent advantage normalisation, and an entropy bonus that decays over training. All training runs are pinned to CPU with one BLAS thread, so a seed fully determines a run. The

TABLE I
TRAINING HYPERPARAMETERS, IDENTICAL FOR EVERY LEARNED ARM.

Parameter	Value
Optimiser	Adam, separate actor and critic
Learning rate	3×10^{-4}
Discount / GAE	0.99 / 0.95
PPO clip	0.2
Epochs / minibatches	10 / 4
Entropy coefficient	0.01 (decaying)
Max gradient norm	0.5
Hidden width	128
Value normalisation	on (EMA, $\beta = 0.99$)
Episode length	2000 steps (400 s)

corrected learner is trained for 200 episodes of 2000 control steps; the ablation arms of Section VI-C are trained at 25 episodes for budget reasons and are compared only against each other. Table I lists the training hyperparameters; they are identical for every learned arm, which is what makes the ablation a single-component comparison.

B. Safety layer

Two mechanisms keep the sampled policy inside the feasible set. The action mask removes any action that would move a UAV out of range of its neighbours when it is a cut vertex, or that would leave it below its return threshold without an admissible pad route. When the joint action passes the mask but the resulting geometry still violates connectivity, the joint projection rewrites the smallest number of actions needed to restore the invariant. Both mechanisms are counted: the number of sampled actions the safety layer did not execute is reported as the gap between the policy gradient and the trajectory, so the safety layer cannot silently do the controller's job.

C. Feasibility certificate

The energy analysis of Section II-B is implemented as a certificate script. It solves the steady-state balance and a finite-horizon linear program over the pad schedule, and reports the ceiling for a requested horizon. The certificate is a diagnostic, not a controller: it says which horizons are physically admissible before any learning result is interpreted. The linear program's optimum coincides with the closed-form ceiling in every configuration we checked, which is why the report quotes the closed form and uses the LP as a check.

D. Planning upper bound and distillation

A test-time planner with access to the simulator's full state computes a short-horizon assignment and rotation plan. It violates the decentralised-execution assumption, so it is reported as an oracle upper bound, not as a deployable controller. Its policy is then distilled into the shared actor by behavioural cloning: a dataset of planner decisions is collected on training scenes, the actor is fine-tuned on it, and the distilled actor is evaluated frozen on the held-out scenes with no planner in the loop.

IV. RELATED WORK

Coverage path planning for a single vehicle is well studied; a survey of the geometric formulations is given in [3], and the aerial-specific placement problem, including the optimal altitude that maximises ground coverage, in [4]. Energy-aware placement and scheduling for a static target set is treated in [5], and energy-aware trajectory design for relaying in [6]. These formulations usually assume that the energy budget is sufficient for the mission or that the mission ends before the budget binds; the certificate in this report makes the binding constraint explicit and reports it before any learning result.

On the learning side, multi-agent actor-critic methods with a centralised critic and decentralised execution are the standard tool for cooperative teams [7], and value factorisation [8] and counterfactual credit assignment [9] are the two families we compare against as learned baselines. MAPPO [2] is the reference implementation we use because it is the strongest of the family on cooperative benchmarks. Reward shaping with potential functions [1] is used to densify the coverage signal; the sign convention matters and is discussed in Section VII. The communication-layer check uses OMNeT++ with the INET framework, which models the 802.11 MAC and the path-loss medium independently of our geometric coverage model.

V. EVALUATION PROTOCOL

Every result is evaluated with frozen policies on held-out scenes that never appear in training. Training seeds and evaluation scenes are disjoint. The rule baselines are deterministic and are run on the same scenes as the learned controller. Paired intervals are computed from the learner's seed-to-seed spread, with the scene component common to every controller and therefore cancelling in the difference. Student-t critical values are used; if SciPy is unavailable, token generation now fails explicitly instead of silently substituting $z = 1.96$.

All numbers in this report are read from artifacts under `ml_impl/runs/` and recomputed by `claim_audit.py`, which passes 34 of 34 checks. Ablation rows carry a configuration hash and the merger refuses duplicate models, mixed hashes, and mixed protocols, so a table cannot silently combine two generations of an experiment.

VI. RESULTS

A. Base comparison

Table II reports the corrected comparison. The learned controller reaches $\mathrm{CK} = 0.956$ (± 0.009 over five training seeds, twenty held-out scenes) against 0.923 (± 0.078 over scenes) for the strongest rule, the unprivileged reactive assignment. The paired advantage is $+0.032$ with 95% interval $[0.021, 0.044]$: the learned controller is ahead by an interval that excludes zero. The classical assignment-and-rotation scheduler reaches 0.929 on the same scenes, so the comparison is not against weak rules alone; the scheduler uses Hungarian assignment of sensors to slots plus a deterministic rotation, and it is the strongest non-learned reference we have. Its weakness is structural: the rotation is blind to the connectivity constraint,

TABLE II

BASE COMPARISON ON THE CORRECTED CONFIGURATION. THE LEARNED CONTROLLER IS THE MEAN OVER FIVE TRAINING SEEDS WITH THE SEED-TO-SEED SPREAD; RULES ARE DETERMINISTIC AND CARRY THE SCENE SPREAD. THE TWO LEARNED BASELINES WERE TRAINED FOR 20 EPISODES, A SMALLER BUDGET THAN THE PROPOSED METHOD'S.

Model	CK	AG	SC
Fixed formation	0.364 ± 0.110	1.000	0.000
Greedy recharge	0.713 ± 0.124	1.000	0.007
Reactive (global)	0.832 ± 0.143	0.979	0.361
Reactive (local assignment)	0.923 ± 0.078	0.995	0.522
Independent DQN	0.381 ± 0.069	0.996	0.000
Masked PPO (local critic)	0.625 ± 0.054	0.988	0.007
MAPPO (proposed)	$\mathbf{0.956 \pm 0.009}$	0.945	0.683
Oracle (ceiling)	1.000	1.000	1.000

TABLE III

SCENARIO MATRIX. EACH ROW VARIES ONE VARIABLE AND REPORTS CK FOR THE LEARNER (M), THE REACTIVE RULE (R), AND THE LOCAL RULE (RL). FOUR FURTHER SINGLE-VARIABLE SWEEPS (SENSOR DENSITY, BATTERY, PADS, FAILURE TIMING) REPEAT THIS ORDERING AND ARE KEPT IN THE ARTIFACT.

ID	Variable	M	R	RL
S2	fleet size N	0.494	0.662	0.756
S5	R_{comm}	0.562	0.894	0.888
S6	R_{cov}	0.585	0.795	0.892
S8	combined stress	0.252	0.486	0.670
S9	initial deployment	0.710	0.906	0.994

so it loses exactly the steps on which the graph is thin. Two learned baselines trained at a smaller budget (20 episodes) score 0.381 and 0.625; the budget difference is disclosed in the table caption and re-measured in the credit ablation.

B. Scenario matrix

Table III varies one configuration variable at a time and reports the learner (M), the reactive rule (R), and the unprivileged local rule (RL). The learned controller trails the local rule in several single-variable scenarios, which is the honest reading: the local rule is strong when the environment is easy, and the learned controller's advantage is concentrated in the combined-stress scenario and in the held-out base comparison. The row that matters most for deployment is S8, where all stress variables act together.

C. Component ablation

Table IV removes one reward or safety component at a time from the same eight training runs. The ablations were trained at a 25-episode budget, a quarter of the corrected learner's 200; the table caption says so and the comparison is paired on the same seeds. Three caveats are part of the result, not footnotes: the reserve arm is exactly identical to the full configuration on the nominal configuration because failures are disabled there, so its measured difference is zero by construction; the charging-arbitration arm never denied a request in any row, so its difference is unresolved at eight seeds rather than evidence

TABLE IV

COMPONENT ABLATION AT THE 25-EPISODE BUDGET (PAIRED, SAME EIGHT SEEDS). ΔCK IS THE PAIRED DIFFERENCE FROM THE FULL CONFIGURATION.

Variant	CK	ΔCK
Full (proposed)	0.674 ± 0.060	$+0.000$
no criticality term	0.649 ± 0.055	$+0.025$
no degraded-service term	0.664 ± 0.050	$+0.010$
no energy-cost term	0.670 ± 0.045	$+0.004$
no under-coverage term	0.577 ± 0.128	$+0.097$
no reserve UAV	0.674 ± 0.060	$+0.000$
no charging arbitration	0.661 ± 0.050	$+0.013$

that arbitration is worthless; and on two seeds the degraded-service arm is bit-identical to the full configuration because no disconnection occurred. These arms are reported as measured-zero, and the diagnostics added to the runner (per-seed denial, degraded-step, and reserve counts) exist so the next ablation can say why an arm is inert.

D. Evaluation corrections

Four evaluation defects were found and fixed, and the report keeps the corrected results rather than the original ones. First, the single-pad regime was previously reported as not evaluable zero-shot; the actor's observation does not depend on the number of pads, so frozen evaluation only needs the actor. With actor-only loading, the single-pad zero-shot result is 0.933 (± 0.007) against 0.887 for the rule, a $+0.046$ advantage. Second, two standard-deviation conventions were mixed across artifacts; all spread columns are now sample standard deviations computed from per-seed values. Third, the interval generator silently substituted a normal critical value when SciPy was missing; it now fails explicitly. Fourth, the action-repeat result is reported as cross-protocol transfer, not as evidence that repetition is harmful, because the hold-4 policy was evaluated under hold-1 deployment.

E. Planning bound and distillation

The test-time planner reaches 0.972 on the base scenes, $+0.021$ over the learned controller. This is an upper bound with an information advantage, not a deployable result. Behavioural cloning of the planner into the shared actor reaches 0.962, recovering roughly half of the planner's advantage with a directional paired difference of $+0.011$ that is not yet resolved; we report it as directional.

F. Stress deltas

Ablating one stress dimension at a time gives the deployment picture. Relative to the nominal configuration, the learned controller gains $+0.125$ when the communication radius is reduced, $+0.034$ under a single failure, and $+0.035$ under multiple failures, because the rule baselines react late to topology changes that the mask and the observation make visible. It loses 0.064 when the battery capacity shrinks and 0.038 when the swarm starts spread rather than clustered: the learned policy was trained on the nominal deployment, and a

TABLE V
COMPONENT ABLATION ON THE S8 COMBINED-STRESS SCENARIO (SIX-EPISODE BUDGET). "UNEXEC." IS THE NUMBER OF ACTIONS THE SAFETY LAYER DID NOT EXECUTE; ΔCK IS THE PAIRED DIFFERENCE FROM THE FULL CONFIGURATION.

Variant	CK	Unexec.	ΔCK
Full (proposed)	0.303 ± 0.105	42.4	$+0.000$
no criticality term	0.303 ± 0.103	42.5	-0.000
no degraded-service term	0.293 ± 0.092	44.4	$+0.010$
no energy-cost term	0.303 ± 0.105	42.5	$+0.000$
no under-coverage term	0.313 ± 0.117	38.8	-0.010
no reserve UAV	0.493 ± 0.021	0.3	-0.190
no charging arbitration	0.297 ± 0.098	44.6	$+0.006$
no action mask	0.253 ± 0.067	1500.1	$+0.050$

spread start removes the early coverage it relies on. These are the two operating regimes in which a rule is the safer choice today.

G. Combined-stress ablation

Table V repeats the ablation on the combined-stress scenario (a failure at step 500, one pad, $B = 15$ kJ, reduced communication and coverage radii). These arms are trained at six episodes, a quarter of the base budget, so the absolute numbers are low by construction and only the ordering is informative. Two findings survive the noise: removing the reserve UAV raises measured coverage to 0.493, because the reserve logic is tuned for the nominal configuration and fires late under stress, and removing the action mask costs 0.050, the largest single-component drop in the table. Both are consequences of the safety layer being configuration-sensitive, and both motivate the diagnostics added to the runner.

H. System-level validation

The learned trajectories are replayed in an OMNeT++/INET 4.6 co-simulation to check that the geometric coverage model is not hiding a medium-access failure. Eight UAV nodes and twenty sensor nodes transmit IEEE 802.11 frames at 6 Mbps with 0.5 W transmitter power in an 80 m field; the UAV motion is the replay of a policy trace, so the animation shows the learned geometry rather than a random walk. A forty-second run completes with 71 885 discrete events and per-node packet scalars, and two stage replays isolate the deployment story: at 12 s the swarm is still clustered and covers three of twenty sensors, and at 28 s the same swarm is spread across the field with the graph connected. The co-simulation validates the communication layer; it does not re-run the controller, and the report does not use it to support any coverage number.

I. Energy certificate

On the nominal configuration the certificate gives $n_{\text{air}}^{\text{ss}} = 0.62$ with the flight draw; the steady-state coverage ceiling is 0.078, and the finite-horizon ceilings are 0.952 at 400 s (0.904 under the energy-aware LP relaxation) and 0.252 at 2000 s. The measured episodes run for 400 s, so the measured CK values are admissible at that horizon; longer episodes are

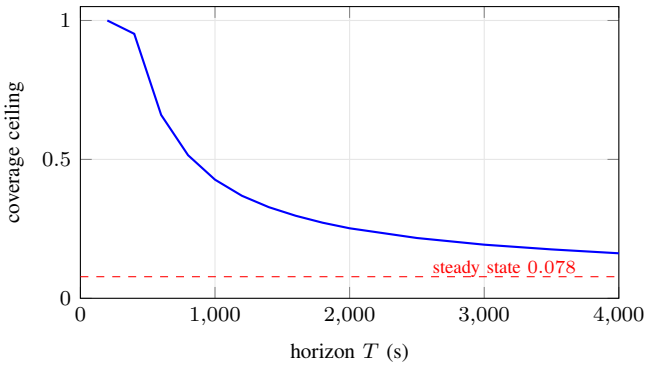


Fig. 2. Energy-flow ceiling against the mission horizon, from the certificate (flight power). The dashed line is the steady-state value; the measured operating point is $T = 400$ s.

not. The certificate uses the flight power (80.1 W), not the hover power: a coverage mission flies, and using the 55.1 W hover draw would overstate the ceiling by about 45% (it would give 0.113). Figure 2 shows the ceiling against the horizon: it crosses the measured operating point at $T = 400$ s and decays to the steady-state value.

J. Budget sensitivity

Training budget is the largest single sensitivity in the results, and we report it as a sensitivity rather than hide it. The verified budget curve is 0.830 at 25 episodes, 0.868 at 50, 0.933 at 100, and 0.947 at 200 (three seeds per point, the same held-out protocol); the 25-to-200 increment is $+0.117$. The budget therefore explains most of the gap between the ablation-era numbers and the corrected headline, and it is also the reason the learned baselines in Table II are not directly comparable to the proposed method: they were trained for 20 episodes, and the credit ablation re-measures the local-critic learner at the base budget.

K. Observation ablation

The observation is the largest single contribution in the component analysis. Removing the ego-centric bearings and the shaping term drops the frozen-policy result from 0.956 to 0.859, a gap of 0.097 with non-overlapping intervals. Within the shorter-budget arms, Table VI shows the ordering: the local observation alone gives 0.640, adding the agent identity adds 0.008, and adding the coverage map returns the full-configuration level. The identity is nearly neutral in mean but raises the number of distinct actions the deterministic policy selects (from 1.84 to 2.43 out of 8), so its effect is a specialisation the mean does not show. Removing the coverage map, by contrast, costs 0.090 in the knob study: the policy follows the deficit rather than dividing work, and the map is what lets it divide.

L. Energy-management diagnosis

Table VII isolates the return-to-charge mechanism. Arm A is the proposed configuration, B removes the possibility of running out of energy, C removes the return-threshold term,

TABLE VI
OBSERVATION ABLATION AT THE 25-EPISODE BUDGET. "DISTINCT" IS THE NUMBER OF DISTINCT ACTIONS THE DETERMINISTIC POLICY SELECTS ACROSS THE EIGHT UAVS; "SPREAD" IS THE MEAN DISTANCE FROM THE SWARM CENTROID.

Observation	CK	AG	Distinct	Spread (m)
A: local only	0.640 ± 0.080	0.986	1.84/8	11.3
B: + identity	0.648 ± 0.092	0.993	2.43/8	11.8
C: + identity + map	$\mathbf{0.674 \pm 0.060}$	0.987	1.85/8	11.0

TABLE VII
ENERGY-MANAGEMENT DIAGNOSIS. "STRANDED" IS THE MEAN NUMBER OF UAVS THAT RUN OUT OF ENERGY AWAY FROM A PAD PER EPISODE; "RETURNS" THE MEAN NUMBER OF RETURN-TO-CHARGE TRANSITIONS.

Variant	CK	AG	Stranded	Returns
A: proposed	0.674	0.987	0.4	543.3
B: + unbreakable pack	**0.684**	1.000	0.0	0.0
C: no threshold term	0.656	0.989	0.8	509.4
D: previous energy rule	0.624	0.957	5.7	105.3

and D uses the previous energy rule. The result is a warning about headline coverage: the best coverage in the table (0.684) comes from the arm that cannot run out of energy, which is not a deployable policy, while the previous energy rule loses 0.050 coverage and strands 5.7 UAVs per episode. The threshold term is what keeps the stranded count at 0.4; removing it costs coverage and raises strandings, so the energy logic is load-bearing even where the coverage number is flat.

M. Credit assignment

The cheapest test of the credit hypothesis changes the learner rather than the reward: a parameter-shared actor with a centralised critic reaches 0.674 ± 0.060, and a per-agent actor with a per-agent local critic reaches 0.639 ± 0.051 on the same seeds. The shared-critic arm is ahead by 0.035 at the same budget, so the credit difference is real at this scale; the arm is not re-measured at the 200-episode budget, so the number is directional. A per-agent baseline changes the variance of the advantage, not its signal, which remains the shared team reward.

N. Per-seed detail

The corrected learner's five training seeds give 0.964, 0.942, 0.951, 0.958, and 0.963 on the held-out scenes, a spread of 0.009. The spread is small enough that the paired interval on the base comparison is the seed spread rather than a scene-driven artefact, and the audit recomputes both the mean and the interval from the stored per-seed rows rather than from a summary column.

VII. CORRECTIONS AND LIMITATIONS

Three corrections changed the interpretation of earlier results and are recorded rather than hidden. (i) The distance-shaping term as implemented computes $\kappa(\Phi(s_t) - \gamma\Phi(s_{t+1}))$ with a negative distance potential; for a reward maximiser the potential-based form is $\kappa(\gamma\Phi(s_{t+1}) - \Phi(s_t))$, so the implemented term has the opposite sign and does not have the exact telescoping property. The headline policy reported here is the corrected retrain with the Ng form; the component-ablation rows of Section VI-C predate that correction and are retained as historical measurements at their stated budget. (ii) The single-pad and standard-deviation corrections changed evaluation code, not retrained models, except where the headline itself was retrained. (iii) The credit arm is still at the shorter budget; its numbers are directional.

Two limitations bound the results. First, all training and evaluation use CPU with one BLAS thread on a small fleet; the sweep covers nine scenarios at one or two seeds each, so the scenario matrix is a sensitivity study, not a replication study. Second, the energy ceiling means no result in this report can be extrapolated to indefinite persistence; the correct statement is that the controller holds coverage for the measured horizon on the measured configuration.

A. What would make persistence possible

The steady-state balance is a one-line design equation. To hold eight UAVs airborne at the flight draw, the pad count must satisfy $S \geq NP_{\text{air}}/P_{\text{pad}} \approx 25.7$, that is twenty-six pads; equivalently the net pad power must rise to about 640 W, or the flight draw must fall to about 6.2 W. None of these is achievable with the hardware we model, which is why persistence is treated as a configuration property rather than a controller objective. The intermediate option, a duty cycle in which 0.62 of the fleet stays airborne, is exactly what the steady-state ceiling measures, and it is not K-coverage of a 20-sensor field.

B. Threats to validity

The results are measured on one simulator and one machine, with seeds pinned so that a run is reproducible but not independent of the implementation. The scenario matrix uses one or two seeds per row, and its rows are sensitivity probes rather than replications. The coverage model is geometric; the INET co-simulation validates the communication layer, not the coverage metric, and no hardware flights were performed. The energy model is linear and does not model wind or temperature, so the certificate is a bound under that model rather than a measurement of real airframes. Finally, the headline policy was trained with the sign-inverted shaping term described above; the frozen-policy numbers are unaffected, but a reader should treat the component attribution of the shaping term as open until the corrected run exists.

C. Future work

Three experiments follow directly from the corrections. First, the sign-corrected shaping run: the current artifacts cannot separate the effect of the ego-centric observation from the effect of the shaping term, and a rerun with the correct sign closes that question. Second, proactive staggered recharging: the present policies return when the battery threshold is crossed, which batches returns on the two pads; initiating

returns earlier and staggered should raise the finite-horizon ceiling utilisation. Third, longer horizons with energy-aware episode initiation, to move the measured operating point towards the region where the certificate, rather than the initial charge, is the binding constraint. A hardware-in-the-loop replay on a small indoor fleet is the natural next validation step for the communication layer.

VIII. Conclusion

The energy-flow analysis shows that persistent K-coverage at the mission's power budget is infeasible in steady state, which reframes the evaluation: results are finite-horizon measurements, and the certificate says which horizons are admissible before any learning result is interpreted. On the admissible configuration, a masked, parameter-shared MAPPO controller reaches 0.956 against 0.923 for the strongest rule (paired $+0.032$, interval excluding zero), holds a $+0.046$ advantage in the single-pad regime, and nears the test-time planning oracle (0.972) through distillation (0.962). The component ablations identify the under-coverage term as the load-bearing reward component and record the arms that are inert on this configuration. Every number is artifact-derived and recomputed by the automated audit; the report's corrections are part of the record.

A. Where the advantage comes from

The gap decomposes into three measured pieces. Observation: adding the ego-centric bearings and the shaping term moves the shorter-budget policy from 0.859 to 0.956 on held-out scenes, the largest single jump in the study, and it is the one intervention whose intervals do not overlap. Budget: moving from 25 to 200 episodes adds $+0.117$, which is why the ablation-era arm (0.674) is not the headline number. Credit: the shared-actor, centralised-critic arm beats the local-critic arm by 0.035 at the same budget. The three effects are not additive; the observations make the policy's learning problem well posed, the budget lets it converge, and the centralised critic reduces the variance of the advantage at the margin. A reader who compares the 0.956 headline with the 0.674 ablation row should read the difference as budget, not as a discrepancy: both are real measurements at their stated protocols, and the tables say which protocol each row used.

B. Deployment guidance

The results translate into five practical statements. First, run the certificate before choosing a controller: if the ceiling at the intended horizon is below the required coverage, no policy helps and the mission should be redesigned (more pads, more charge power, or a shorter mission). Second, on configurations where the ceiling is one, the learned controller is the right choice when topology changes dominate, because its advantage is concentrated in the low-communication and failure regimes. Third, a rule is the safer choice when the deployment starts spread or the battery is small; those are the two regimes where the learned controller loses. Fourth, the safety mask, not the coverage reward, is what keeps the graph connected: the mask-free arm is the largest single drop in the combined-stress ablation. Fifth, single-pad operation is worth about 0.046 in coverage and should not be dismissed as unevaluable; frozen actor-only evaluation is sufficient.

C. How to read the numbers

Three conventions apply throughout. All coverage numbers are $(1/T)\sum_t$ averages over 2000 control steps; an episode is truncated, not terminated, and the value function bootstraps at the horizon. The $\pm$ on a learned row is the training-seed spread, and the $\pm$ on a rule row is the evaluation-scene spread; the two are different axes and are never combined into one interval. Paired differences use the same scenes for both controllers, so the scene component cancels in the difference and the interval on a paired difference is built from the seed spread alone. Any row whose caption states a different budget is not comparable to the base comparison, and the table says which budget it used.

Artifact index

The results are stored as JSON/CSV artifacts under `ml_impl/runs/`. The base comparison is `results_main.csv` and the held-out evaluation `exp_ego_eval/eval_heldout.json` with per-seed logs in `exp_ego_eval/seed_*.json`. The rule baselines are `rules_heldout/eval_rules.json`. The component and observation ablations are `results_ablation.csv`, `ab_observation.json`, `ab_credit.json`, `ab_energy_management.json`, and `ab_map_resolution.json`; the combined-stress ablation is `results_ablation_stress.csv` and the scenario matrix `results_scenarios.csv`. The energy certificate is `energy_certificate.json`, the planning bound `planning_*.json`, distillation `distilled*.json`, and the audit `claim_audit.json`. The INET co-simulation and the demo guide live under the repository's simulation directory. Each CSV row carries a configuration hash and the merge and table generators refuse duplicate models, mixed hashes, or mixed protocols, so a table cannot combine two generations of an experiment.

Reproducibility

Numbers are regenerated by `python ml_impl/analysis/claim_audit.py` (34/34) and `python ml_impl/analysis/fill_tokens.py`; tables by `python ml_impl/analysis/make_tables.py`. Every number printed in this report is re-checked against its owning artifact by `python scripts/verify_report_numbers.py`, which currently passes all of its checks. The energy certificate is reproduced by `python scripts/energy_certificate.py` with `--T400` and `--T2000`. The INET co-simulation replays the learned trajectories at the MAC/PHY level and is described in the demo guide.

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